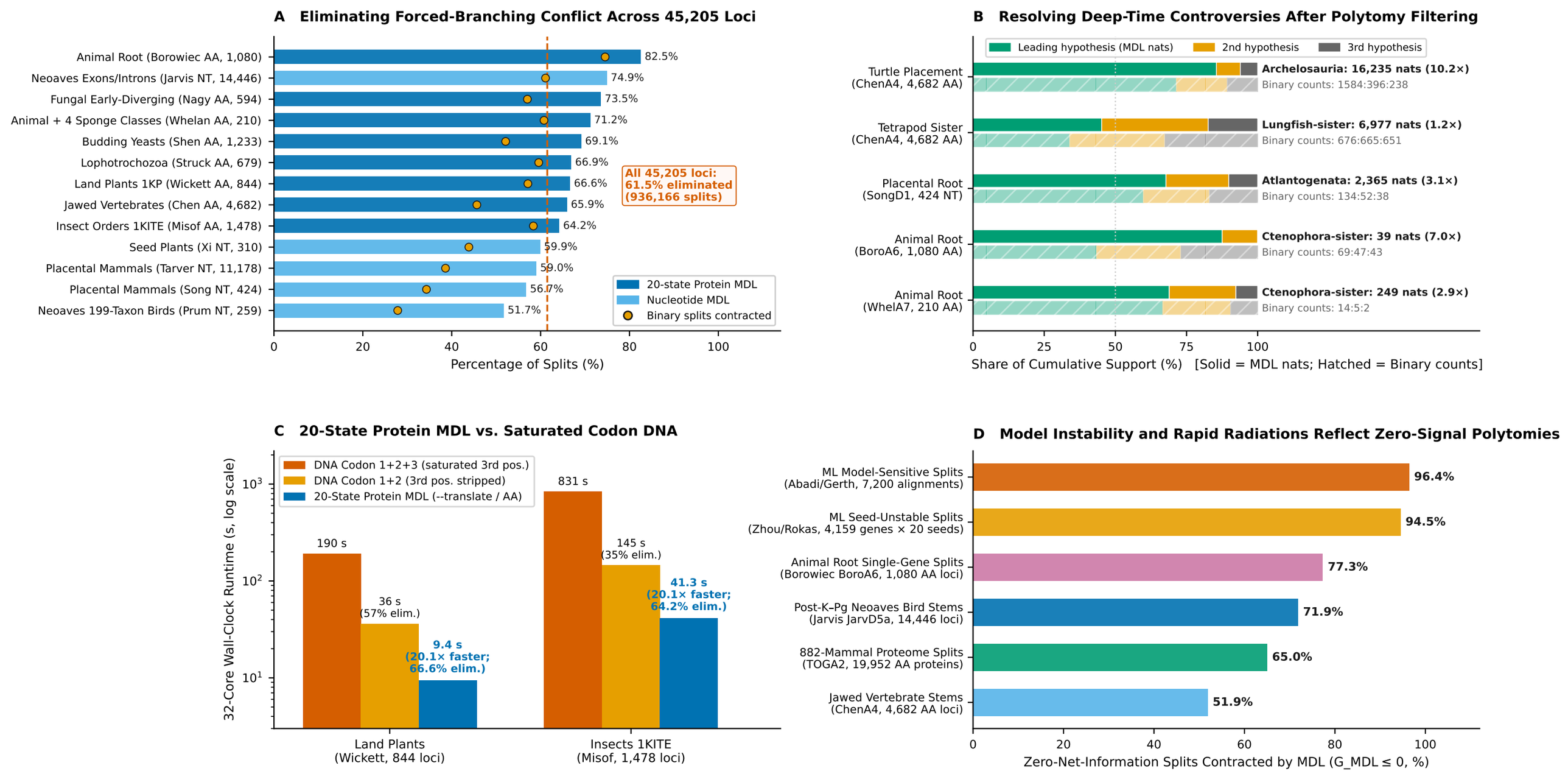
A Eliminating Forced-Branching Conflict Across 45,205 Loci
Animal Root (Borowiec AA, 1,080)
Neoaves Exons/Introns (Jarvis NT, 14,446)
Fungal Early-Diverging (Nagy AA, 594)
Animal + 4 Sponge Classes (Whelan AA, 210)
Budding Yeasts (Shen AA, 1,233)
Lophotrochozoa (Struck AA, 679)
Land Plants 1KP (Wickett AA, 844)
Jawed Vertebrates (Chen AA, 4,682)
Insect Orders 1KITE (Misof AA, 1,478)
Seed Plants (Xi NT, 310)
Placental Mammals (Tarver NT, 11,178)
Placental Mammals (Song NT, 424)
Neoaves 199-Taxon Birds (Prum NT, 259)
82.5%
74.9%
73.5%
71.2%
69.1%
66.9%
66.6%
65.9%
64.2%
59.9%
59.0%
56.7%
51.7%
All 45,205 loci:
61.5% eliminated
(936,166 splits)
20-state Protein MDL
Nucleotide MDL
Binary splits contracted
0
20
40
60
80
100
Percentage of Splits (%)
B Resolving Deep-Time Controversies After Polytomy Filtering
Leading hypothesis (MDL nats)
2nd hypothesis
3rd hypothesis
Turtle Placement
(ChenA4, 4,682 AA)
Archelosauria: 16,235 nats (10.2×)
Binary counts: 1584:396:238
Tetrapod Sister
(ChenA4, 4,682 AA)
Lungfish-sister: 6,977 nats (1.2×)
Binary counts: 676:665:651
Placental Root
(SongD1, 424 NT)
Atlantogenata: 2,365 nats (3.1×)
Binary counts: 134:52:38
Animal Root
(BoroA6, 1,080 AA)
Ctenophora-sister: 39 nats (7.0×)
Binary counts: 69:47:43
Animal Root
(WhelA7, 210 AA)
Ctenophora-sister: 249 nats (2.9×)
Binary counts: 14:5:2
0
25
50
75
100
Share of Cumulative Support (%) [Solid = MDL nats; Hatched = Binary counts]
C 20-State Protein MDL vs. Saturated Codon DNA
DNA Codon 1+2+3 (saturated 3rd pos.)
DNA Codon 1+2 (3rd pos. stripped)
20-State Protein MDL (--translate / AA)
32-Core Wall-Clock Runtime (s, log scale)
10^3
10^2
10^1
190 s
36 s
(57% elim.)
9.4 s
(20.1× faster;
66.6% elim.)
831 s
145 s
(35% elim.)
41.3 s
(20.1× faster;
64.2% elim.)
Land Plants
(Wickett, 844 loci)
Insects 1KITE
(Misof, 1,478 loci)
D Model Instability and Rapid Radiations Reflect Zero-Signal Polytomies
ML Model-Sensitive Splits
(Abadi/Gerth, 7,200 alignments)
96.4%
ML Seed-Unstable Splits
(Zhou/Rokas, 4,159 genes × 20 seeds)
94.5%
Animal Root Single-Gene Splits
(Borowiec BoroA6, 1,080 AA loci)
77.3%
Post-K–Pg Neoaves Bird Stems
(Jarvis JarvD5a, 14,446 loci)
71.9%
882-Mammal Proteome Splits
(TOGA2, 19,952 AA proteins)
65.0%
Jawed Vertebrate Stems
(ChenA4, 4,682 AA loci)
51.9%
0
20
40
60
80
100
Zero-Net-Information Splits Contracted by MDL (G_MDL ≤ 0, %)